

Approximating spatial processes with too many knots degrades predictive accuracy of SPDE models

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Abstract

1. Spatial and spatiotemporal models are seeing increasingly used in ecology and related fields to model latent distributions. Applications relying on these methods have used them to track population change over time, assess changing species distributions, and model spatial population processes (predation, habitat use, fishing impacts, etc).
2. Computational advances over the last decade have allowed for the rapid estimation of spatial approximations to high dimensional spatial surfaces (100s or more georeferenced points), and these methods have been accessible in software implementing the Integrated Nested Laplace Approximation (INLA). One of the most critical decisions for building INLA-based models is how to construct the spatial approximation (“mesh”). Little guidance has been established in constructing these approximations, with many previous applications suggesting that higher resolution meshes result in better performance. Using a publicly available dataset of commercially fished species on the west coast of the USA, we develop two cases studies; first we construct a spatial model of ocean temperature, and second we build spatial models of species occurrence and catch rates for 6 species. We apply a series of sensitivity analyses to each example to understand how mesh resolution affects the predictive accuracy of models predicting species occurrences and catch rates.

3. Our results highlight that for spatial models of temperature and catch rates, the finest scale spatial approximations generally overfit the data and resulted in decreased out of sample predictive performance. For each of the species and models considered, there appeared to be an optimal range of vertices (knots) used in constructing the spatial approximation. These results held across a range of spatial cross-validation sampling schemes. We found less clear results when working with binomial presence absence data – while finer mesh resolutions resulted in better fits to training data,
4. Results from our work highlight that practitioners should not assume that the performance of SPDE-based models gets better with higher resolution. Instead, each dataset likely has a range of resolutions that results in optimal out of sample predictive performance; the resolution of these meshes may be identified using spatial cross-validation approaches, such as the sampling procedures used in our analysis. Selecting models with the appropriate spatial complexity is expected to improve parameter estimates and derived quantities (estimates of population abundance, distribution, etc.) when out of sample prediction is a focus of inference.

Key words

Spatial, spatiotemporal, Stochastic Partial Differential Equation (SPDE), Integrated Nested Laplace Approximation (INLA), Gaussian random field, mesh, predictive accuracy

Introduction

Recent computational advances have allowed for the rapid estimation of large spatial and spatiotemporal models in ecology and related fields (Banerjee & Fuentes 2012). Motivating questions for these applications to large georeferenced datasets including tracking abundance in time and space, identifying environmental predictors of occurrence, modeling disease spread, or quantifying range shifts in distribution (Ross *et al.* 2012; Miller *et al.* 2013; Thorson & Barnett 2017).

There are a number of statistical approaches for developing models with spatial components; examples include generalized additive models (GAMs) (Murase *et al.* 2009; Forney *et al.* 2012), non-parametric approaches (Mutanga *et al.* 2012), and applications of Gaussian random fields (Latimer *et al.* 2009; Datta *et al.* 2016). Estimating these models can be computationally challenging for large spatial datasets (Banerjee *et al.* 2004), necessitating the use of approximations or smoothing. One solution to this problem is the Stochastic Partial Differentiation Equation (SPDE) approach (Lindgren *et al.* 2011). The SPDE method uses Gaussian Markov Random Fields (GMRF) to model the conditional distributions of values as only dependent on neighboring points; thus, rather than estimating a full rank covariance method to describe the

spatial field, a sparse precision matrix can be estimated (Lindgren *et al.* 2011). Like other approximations, the SPDE approach can be seen as a form of smoothing (Yue *et al.* 2014). (Miller *et al.* 2020) provide an excellent summary of the recent SPDE literature, and also show the link between these approaches and smoothing splines.

Advances by (Rue *et al.* 2009) have enabled the SPDE approach to be implemented with the Integrated Nested Laplace Approximation (INLA) as a fast and efficient approximation to estimating high-dimensional Gaussian random fields. Rather than using MCMC to estimate the posterior distribution of parameters, INLA approximates the posterior with the Laplace approximation. These models have been extended to include the ability to fit geographic barriers, such as islands or lakes, anisotropic covariance functions, and non-Gaussian data (Bakka *et al.* 2018). In addition to estimating Bayesian posterior distributions (Lindgren & Rue 2015), INLA has been integrated with Template Model Builder (TMB; (Kristensen *et al.* 2016; Osgood-Zimmerman & Wakefield 2021)) to enable auto-differentiation and has been adopted in many spatial fisheries applications around the world (Anderson *et al.* 2019a; Thorson 2019b; e.g., Johnson *et al.* 2019).

Constructing spatiotemporal models with INLA or related software involves many of the same considerations as with simpler non-spatial models. For example, modelers frequently ask: “Which covariates are most important?” and “Does my model do a satisfactory job of making predictions?” Spatial or spatiotemporal models may include additional considerations, such as specifying the form of spatiotemporal variation (Anderson & Ward 2019). One of the more complicated and important decisions—particularly for models implementing the SPDE-INLA approach—lies in specifying the dimensionality of the latent spatial fields (Lindgren & Rue 2015). Unlike Generalized Additive Models (GAMs), which often penalize smooth terms (Wood 2003), there are not explicit penalties in overfitting with the SPDE approach in INLA.

Spatial fields are approximated using the SPDE approach by first constructing a “mesh” (a map consisting of small triangles, e.g., Figure 1). By estimating values of the spatial field at the vertices (also “nodes” or “knots”), the spatial field may be triangulated to coordinates where data are observed or predictions are made (Lindgren *et al.* 2011). There are a number of decision points to be made in constructing the mesh, including whether to include a boundary, whether the boundary is regular, the maximum edge of any triangle (in the core or boundary area), and the minimum angles allowed by the triangles (Gómez-Rubio 2020). Each of these decisions can impact the density and number of small triangles that make up the mesh—and in turn, the accuracy of predictions. Though the mesh construction is important, this step has been omitted from many best practices guides using the SPDE-INLA approach, and we are only aware of several sensitivity analyses examining the importance of mesh construction (Righetto *et al.* 2020; Røste 2020). While many previous applications and guidance around INLA have suggested that increasing the resolution of the mesh

improves predictive accuracy (Thorson 2019a; Godana *et al.* 2019; Williamson *et al.* 2021; Wilson 2021), a much smaller number of studies have noted that finer meshes don't always translate into better performance (Huang *et al.* 2017).

The objective of this paper is to examine whether higher resolution spatial approximations in INLA translate to better predictive accuracy, and provide general recommendations for identifying models with good out of sample predictive performance. Using a real-world case study of groundfishes and associated environmental covariates (temperature) in the Northeast Pacific Ocean, we evaluate the out of sample predictive performance as a function of mesh resolution (number of knots). We further evaluate these results across a range of cross-validation sampling schemes (number of spatial blocks and folds), including an extreme case of leave-one-out cross-validation (LOOCV). All data for our analysis are publicly available, and code and data to reproduce our analysis is available on Github (XXXX) and Zenodo (on publication).

Methods

Data

To evaluate the effect of mesh resolution on predictive ability of INLA-based models, we use a publicly available scientific survey of marine fishes in the Northeast Pacific Ocean as a motivating example (<https://www.nwfsc.noaa.gov/data>). The West Coast Bottom Trawl Survey (WCBTS) data represents a fishery-independent survey collected annually over a 16-year period, 2003-2021, covering the California Current ecosystem on the west coast of the USA (Keller *et al.* 2017). The survey is designed to estimate the abundance, sizes, and age composition of commercially and recreationally targeted groundfish species found in near-bottom habitats. The fundamental methods of the survey, sampling intensity, gear, seasonal and spatial coverage, and random stratified design have remained relatively constant. In addition to collecting biological samples, the survey also measures environmental variables, such as water temperature at the depth of the net. Because of the large number of model runs needed to evaluate our sensitivity analyses, we restricted our analysis to only use survey data from 2018 ($n = 702$ observations). Though hundreds of species were encountered in the survey, we further restricted our analysis to six species that are of interest to fisheries, but also represent a range of occurrences (Table S1).

Modeling temperature

As a first example, we construct spatial models of June 2018 ocean temperature at depth; this represents a subset of the 2018 data ($n = 175$ of 702 tows) and unlike biological responses, it is reasonable to assume temperature residuals are normally distributed. As there are generally strong non-linear trends between

depth and temperature, we used a statistical model of the form

$$g(u_s) = b_0 + b_1x + b_2x^2 + \omega_s$$

where ω_s is the estimated spatial field, b_0 is the intercept, x is the depth associated with each observation, b_1 and b_2 are estimated quadratic coefficients, u_s is the expected value for point in space s , and $g()$ is an inverse link function.

We conducted two analyses, using two splits of the data. The observations were first split with 25 observations randomly assigned to the test set (reserving $n = 150$ for training the model). As a second split, we applied true Leave-One-Out Cross-Validation (LOOCV), with each observation alternatively held out as the test set. All of the meshes in our analyses used a non-convex boundary and offset based on 5% of the data (e.g. Figure 1). We fixed the maximum edge of triangles in the mesh at 100km in the core area and 500km closer to the boundary. The mesh resolution was varied by modifying the cutoff distance (representing the distance where two points are treated as a single vertex) because the cutoff distance has been shown to have the greatest impact on measures of model performance from mesh construction (Righetto *et al.* 2020). There is an inverse relationship between cutoff distance and mesh resolution or knots; we varied cutoff distances between 20–160km in increments of 2.5km. We fit all models using R-INLA (Lindgren & Rue 2015; R Core Team 2021) and R 4.1.0 (R Core Team 2021) and specifically the ‘inlabru’ R package, which is a convenient wrapper for the INLA software (Bachl *et al.* 2019). Penalized complexity priors were used on the Matérn covariance parameters (Simpson *et al.* 2015; Osgood-Zimmerman & Wakefield 2021). Specifically, priors on the Matérn spatial range (κ) and marginal standard deviation (σ_ω) are constructed as a function of the tail probabilities (Simpson *et al.* 2017; Fuglstad *et al.* 2018). We assigned priors so that $Pr(\kappa < 20) = 0.05$ and $Pr(\sigma_\omega > 5) = 0.05$.

Quantifying performance

To compare the predictive performance of models with alternative mesh configurations, we considered several out of sample predictive performance. We first focused on the conditional predictive ordinate (CPO, (Pettit 1990)), which is an approximation to the LOOCV. The total predictive log density can be represented as the sum of individual log posterior probabilities, $\sum_{i=1}^n \log[P(Y_i|Y_{\neq i})]$, where $P(Y_i|Y_{\neq i})$ is the probability of test data i . Because conducting true leave-one-out cross-validation may be prohibitive for large datasets, INLA uses CPO to approximate the out of sample posterior probability for each observation. Previous research has

shown that the CPO approximation performs similarly to MCMC cross-validation (Held *et al.* 2010), yet it is less clear whether CPO is similar to true cross-validation in comparing mesh resolution. For comparison, we calculated the total CPO score across a range of cutoff values, and compared these results to the true predictive density using LOOCV. For our comparison using a larger test set (25 observations), we calculated the predictive log density for each observation. Additional metrics, including the bias and standard error associated with each prediction were also used for comparisons.

Modeling species occurrence and catch rates

As a second case study, we focus on modeling fish densities from the WCBTS survey. As data collected by the WCBTS and other marine surveys are both zero-inflated and have positively skewed catch rates, conventional approaches to analyzing these data have used a delta or hurdle model to separately model occurrence and positive catches (Thorson 2019a; e.g., Johnson *et al.* 2019). We adopt a similar approach in our modeling, and apply two sub-models to the data; we use a binomial distribution with a logit link to model species occurrence, and a Gamma distribution with a log link to model positive catch rates (kg per km²). For consistency with recent research using these data (@ Johnson *et al.* 2019), we applied the same model used for bottom temperature; this consisted of an estimated intercept, quadratic effect of depth, and estimated spatial field. Though the WCBTS survey collects data on hundreds of marine fishes, we limited our analysis to 6 species that are of interest to fishery managers, and that represent a range of distribution and occurrence (Table S1; Figs. S1–S2).

As a first sensitivity, we constructed 20 alternative meshes using cutoff values ranging from 3–120km for each combination of species and hurdle sub-model. Meshes varied slightly by species and sub-model, but this range of cutoff values resulted in meshes ranging in size from 75–1348 knots. High-resolution meshes with more knots than observations were included as an upper extreme, though many applications of the SPDE approach to similar fisheries survey data have used 500–600 knots (e.g., Thorson 2019a; Tolimieri *et al.* 2020).

As a second sensitivity, we varied how data were split into training and test sets to evaluate model performance. We used 10-fold cross-validation for all comparison, and considered three alternative splits of the data. Recent work has demonstrated that for spatially structured data, creating random folds that ignore spatial autocorrelation may bias results (Valavi *et al.* 2019). Given the WCBTS survey is nearly 1-dimensional along a latitudinal gradient (Keller *et al.* 2017) (Figure 1), we used latitude quantiles to divide the WCBTS survey domain into blocks ($n = 20, 40$) with folds assigned continuously to blocks. As a third split, we assigned observations randomly to folds, ignoring spatial structure. Additional blocking

strategies, using the ‘blockCV’ package in R (Valavi *et al.* 2019) were also considered (Fig in SI?). Model performance was evaluated by calculating the predictive log density for each fold, using either the binomial or Gamma densities, and total predictive densities were calculated by summing across folds. To contrast the out of sample predictive performance with fit to the training data, we also calculated the predictive density of the training data.

As a third sensitivity, we examined the robustness of the cross-validation results to the choice of hyperparameters for the PC priors (Matérn κ and σ_ω). Because of the dimensionality of parameters to explore across species and sub-models, we only used the Dover Sole presence-absence sub-model. The PC prior on σ_ω is to specify the threshold and upper tail probability, $Pr(\sigma_\omega > \sigma_0) = \theta_\sigma$. In addition to the original values ($\sigma_0 = 5$, $\theta_\sigma = 0.05$), we examined alternative values of $\sigma_0 = (1, 10)$ and $\theta_\sigma = c(0.01, 0.1, 0.4)$. In contrast, the PC prior on κ specifies the lower threshold and tail probability, $Pr(\kappa > \kappa_0) = \theta_\kappa$. We varied κ_0 (5, 20, 40) and θ_κ (0.01, 0.1, 0.4). For each combination of PC priors, we calculated the out of sample predictive density, using 10-fold cross-validation and the same latitudinal bands as in our first sensitivity analysis.

As with our spatial models of ocean temperature, groundfish models were fit using inlabru (Bachl *et al.* 2019). Parameter estimates were also validated by also estimating the same models using TMB (Kristensen *et al.* 2016). Specifically, we performed estimation using the R package ‘sdmTMB’ (Anderson *et al.* 2019b), which uses INLA to create meshes, but takes advantage of autodifferentiation in TMB for faster maximum likelihood parameter estimation.

Results

Temperature modeling

Our spatial models of ocean temperature demonstrated that the relationship between cutoff distance (or vertices in the mesh) and predictive density was generally in opposite directions for data used to fit the model versus out of sample data used to evaluate predictions (Figure 2). While predictive density was highest at small cutoff distances (~ 20 km) corresponding to a fine mesh, we found that a coarser mesh with larger cutoff distance (~ 130 km) had the highest predictive density for the test data. Increasing the mesh resolution generally improves the spatial model point estimates for training data, but a consequence of adding more vertices is that the standard errors of predictions increase for both in-sample and out of sample data (Figures S3-S6). Evaluating the true LOOCV against the CPO metric was largely confirmatory, with both approaches highlighting that intermediate meshes (cutoff distances ~ 80 km) yielded improved out of

sample prediction for this data (Figure S7).

Groundfish modeling

Our models of groundfish presence-absence and catch rates demonstrated that as expected, higher resolution SPDE meshes generally yielded better fits to training data (Figures 3-4). The effect of mesh resolution on the fit to training data was weakest for species with lowest occurrence rates (Pacific Ocean perch, Darkblotched rockfish, Figure 3). The relationship between mesh resolution and out of sample predictive performance appeared more variable. When spatial structure is ignored in designating folds, finer meshes appear to have the best out of sample performance across species, however when spatial blocks are used for cross validation the finest meshes don't consistently outperform coarser ones (Figure 3).

Results for spatial models of species catch rates appear to be more consistent with results from our spatial model of ocean temperature, in that as in sample predictive performance increases with finer mesh resolution, out of sample predictive performance decreases (Figure 4).

with the binomial and Gamma models across a range of species and mesh cutoff values demonstrated that after some optimal spatial cutoff of triangles in the mesh, predictive accuracy of the spatial models deteriorated as mesh resolution was increased. Binomial models of Darkblotched rockfish (Fig. 2) and Gamma models for all species also supported a decline in performance with the coarsest mesh resolutions (Fig. S1). The binomial sub-model suggested that the optimal number of knots was 200-250 (Fig. 2), and the positive Gamma sub-model indicated even smaller knot values had nearly equivalent out of sample predictive performance (Fig. S1). In contrast to the decline in out of sample performance with high mesh resolution, we found the opposite to be true for in sample data; for both the binomial and Gamma models, predictive performance generally increased with the number of knots (Fig. 2, Fig. S1).

Focusing on the Dover Sole presence-absence model, we examined sensitivity to spatial sampling and choice of penalized complexity parameters for the Matérn covariance function. To examine effects of spatial sampling, we adjusted the buffer separating test and training dataset (Fig. 3) and adjusted the number of folds used for cross-validation (10, 40, Fig. S3). There is an inverse relationship between the buffer size and number of blocks, with large buffers (125km, 150km) corresponding to small numbers of blocks for this dataset (25, 19). With these largest buffers, there appears to be a decline in predictive performance at small numbers of knots (Fig. 3, Fig. S3), but with large numbers of spatial blocks (> 50), the coarsest meshes have similar performance as models with moderate numbers of knots. Varying the hyperparameters of the PC priors for the Matérn κ and σ_ω similarly demonstrated that the predictive performance of these models declined across all values (Fig. SX).

Because k-fold cross-validation requires significant computation time, approximations like CPO may be more convenient ways to measure predictive accuracy of spatiotemporal models. Though conventionally used to compare models with alternative fixed effects, we compared CPO approximations to true LOOCV, as an extreme case of cross-validation. Both CPO and LOOCV suggested that too many knots can result in diminished predictive accuracy (Figure 4). The biggest difference in these metrics was that the LOOCV suggested that more than 400 knots resulted in decreased performance, while CPO indicated a performance decline after 675 knots (Figure 4).

Finally, we examined the effects of increasing mesh resolution on estimated model parameters. We found little evidence of patterns between mesh resolution, estimated Matérn parameters (κ , σ_ω), and predictive log densities. While the estimated range (κ) generally declined with large numbers of knots (Fig. 5), there are less obvious patterns in the marginal standard deviation σ_ω (Fig. 5). The estimated fixed effect B_0 representing the mean of the spatial field generally declined with increasing mesh resolution (Figure S2), because as the mesh increases in complexity and explains more variation in observed data, the intercept is expected to shrink toward the mean.

TODO (WORK IN SDMTMB COMPARISON, FIGURE FROM SEAN?)

Discussion

Spatial and spatiotemporal models are becoming increasingly used in ecology, fisheries, and related subfields, in particular to address questions related to changing species distributions in time and space. High dimensional datasets with hundreds of observations introduce computational challenges, but complicated spatial structure can be efficiently approximated using the SPDE approach (Miller *et al.* 2020). Combined with the INLA software for estimation (Rue *et al.* 2009), these tools have become widely accessible. Though a number of examples and best practices papers have been written to demonstrate how to construct models or specify hyperparameters (Lindgren & Rue 2015; Gómez-Rubio 2020), advice on specifying the SPDE mesh to approximate spatial processes can be confusing.

previous work has shown parameter estimates converge as mesh resolution becomes sufficiently large

Identifying the spatial complexity introduces new computational challenges for analysts, but is important for improving predictions and derived quantities of interest (indices of population abundance, area occupied, centre of gravity, etc.).

Using a publicly available dataset of groundfishes in the NE Pacific Ocean, we use a number of sensitivity analyses to demonstrate an important tradeoff between mesh resolution and the predictive accuracy of spatial

models using the SPDE-INLA approach. As expected, models with higher resolution meshes (more knots or vertices) are more flexible and can better explain variation in data used to fit the model. For applications interested in identifying models with good predictive ability for future data (either in space or time), we demonstrate that models with the finest spatial approximations have deteriorating predictive performance for out of sample data. Declines in out of sample predictive performance with too many knots appears robust to (1) how data are spatially sampled for cross-validation, (2) the specific penalized complexity priors used for the Matern function, and (3) whether estimation is done using INLA or maximum likelihood with TMB. Datasets like the groundfish case study used here ($n = 702$) support meshes with ~ 300 or fewer knots, though some of the models supported even fewer knots.

While there are a number of alternative ways to measure model performance, we recommend using k-fold cross-validation to explore predictive accuracy across a range of meshes. Alternative measures of model fit exist, including the conditional predictive ordinate (CPO) (Pettit 1990). While the CPO may approximate LOOCV for some datasets, we found a mismatch between CPO and LOOCV, with CPO supporting meshes with very large numbers of knots and true LOOCV supporting fewer. Other tools like AIC have also been used to discriminate among these types of models, however AIC is known to be problematic when used as a metric to compare mixed effects models (Harrison *et al.* 2018) or as a tool to estimate spatial complexity (Velasco & González-Salazar 2019).

Comparing the performance of models with alternative spatial complexity obviously introduces additional computational burdens, both in performing cross-validation, and comparing models with different numbers of knots. This challenge is not unique to models using the INLA-SPDE approach; selecting both the location and number of knots is a general problem for knot-based spatial models. Other examples of spatial statistics models that have encountered this issue include machine learning methods (reviewed in (Banerjee *et al.* 2012)), Gaussian Process models (Garton *et al.* 2020) and P-splines (Ruppert 2002). Identifying a range of knot values that yield good predictive performance will be specific to individual datasets (e.g. magnitude of observation error, spatial autocorrelation, number of observations, type of response). The optimal mesh resolution for one species may be informative for other species from the same dataset, potentially reducing the computational overhead.

3. Rules of thumb? This is tricky because it's somewhat dataset specific. The blockCV MEE paper suggests defining spatial blocks based on residual spatial autocorrelation – which involves fitting the model, so somewhat circular [and tools like calculating the variogram are made difficult by non-normal response data]. Most advanced GP approaches include knot selection as a parameter, or do an adaptive grid search

The number of knots needed to generate a model with high predictive accuracy depends on the other features of the model. Some of this is based on data (e.g. homogenous spp like Dover sole vs patchily distributed rockfish), but a lot of this is dependent on how much variation is parsed into observation error. High obs errors will result in flatter spatial fields, which will require fewer knots.

4. Future research:

- is this a feature of triangulation with inla? would we get better estimates from a higher dimensional GP model?

– both bias and precision, fig s2 - don't use AIC - identifying the appropriate spatial complexity is important for

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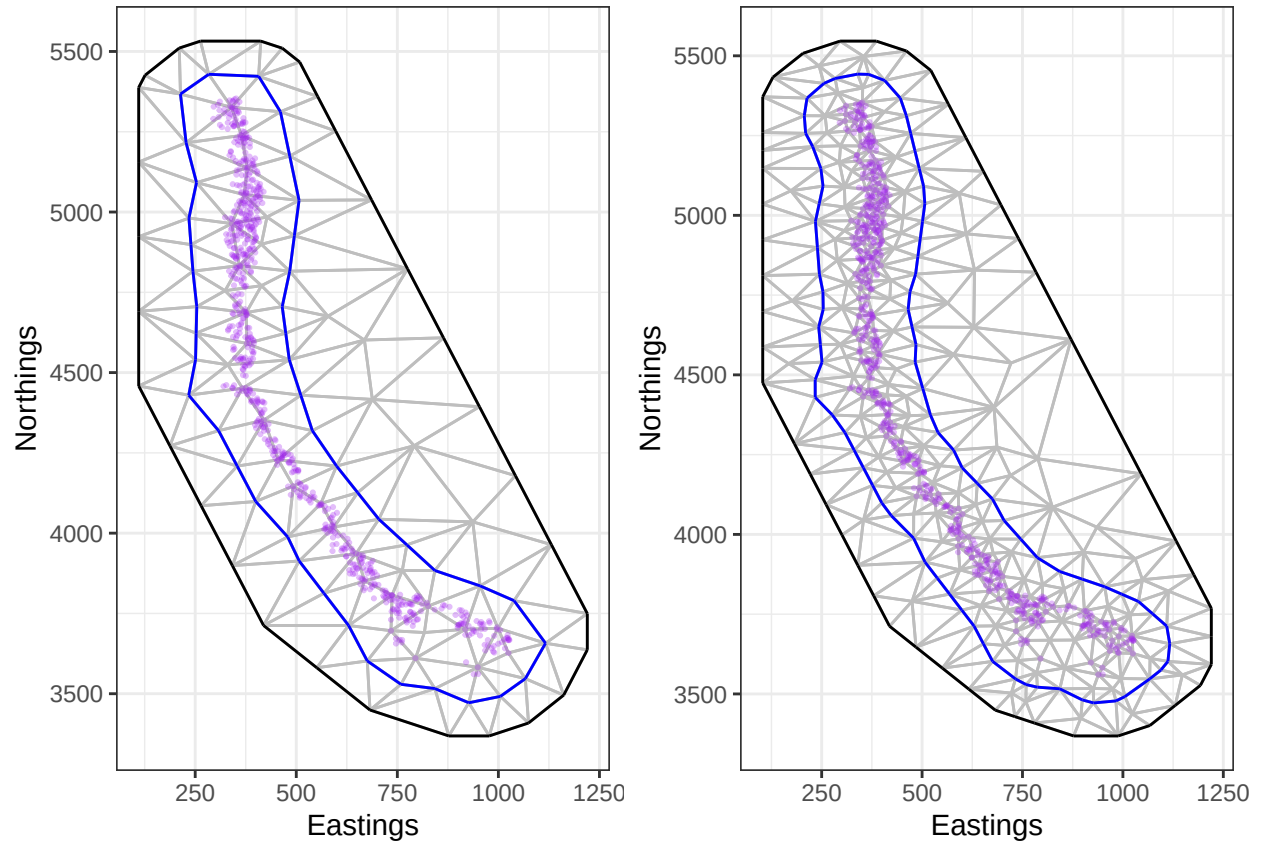


Figure 1: Example of two meshes constructed for the California Current Ecosystem, using the 2018 WCBTS survey data. Cutoff values of 80 and 20km are used, translating into 110 and 394 knots (vertices) respectively.

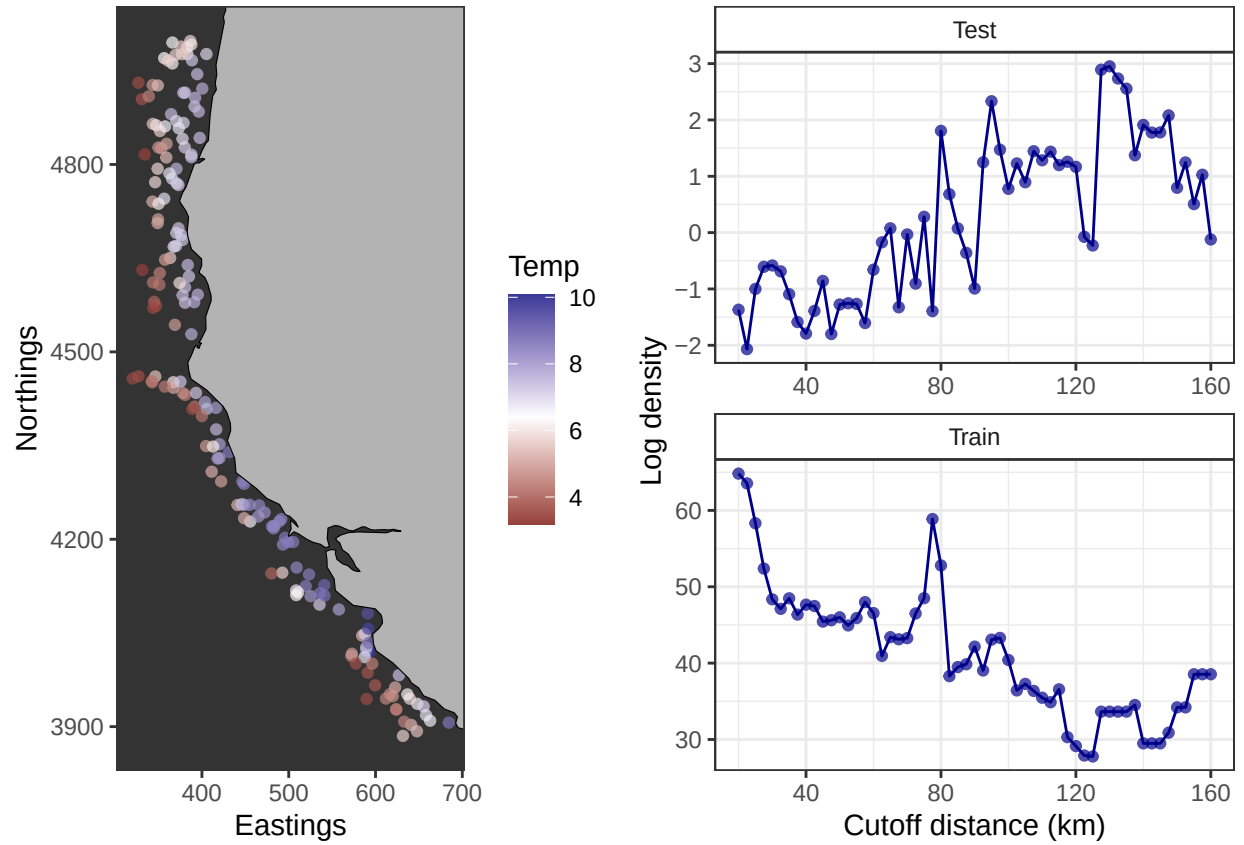


Figure 2: June ocean temperature (Celsius) from the 2018 WCBTS survey (sample locations further offshore are typically at greater depth, and colder). Log predictive densities are shown for the test and training datasets in our modeling. Each predictive density point represents a different mesh, constructed by varying the cutoff distance (x-axis); higher cutoff values correspond to coarser meshes.

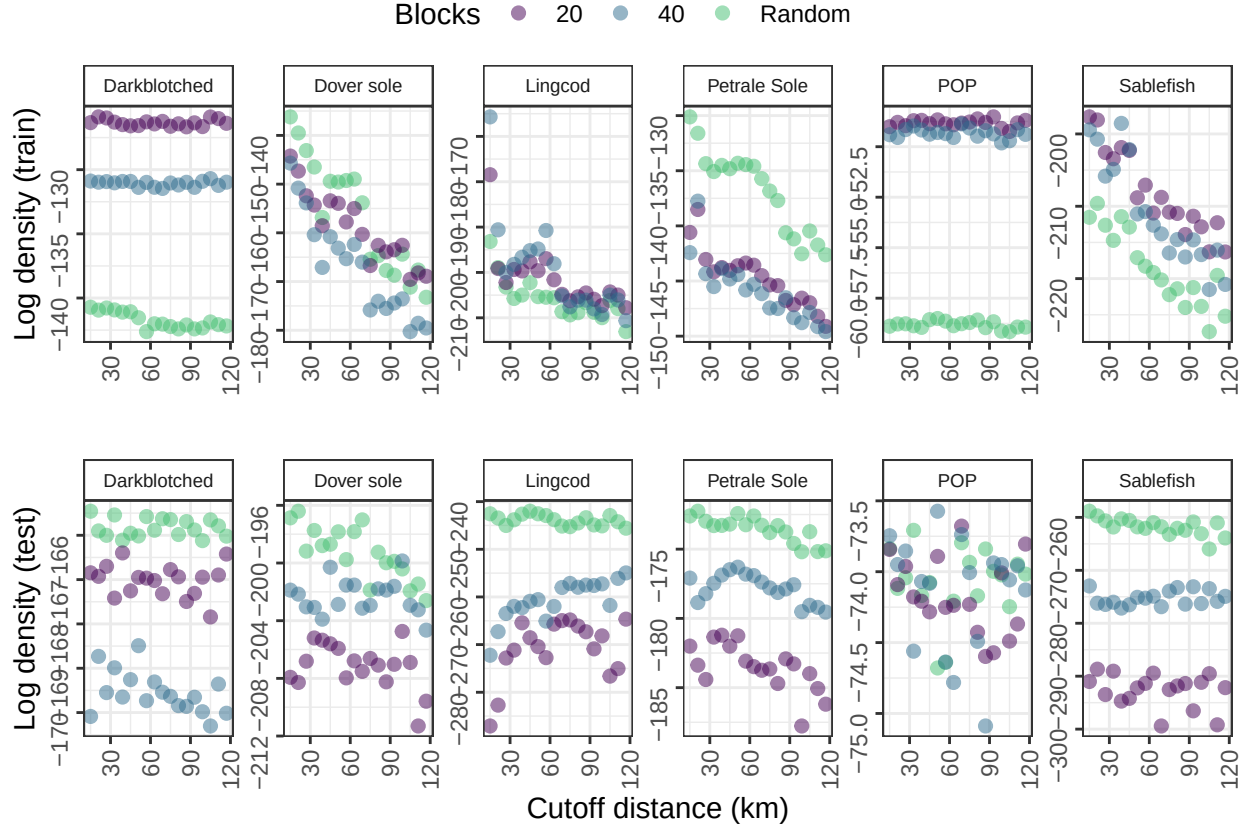


Figure 3: Binomial predictive densities for occurrence models for six species from the WCBTS survey, as a function of cutoff distances (larger cutoff distances correspond to coarser meshes). The in-sample predictive density is calculated holding out the first fold.

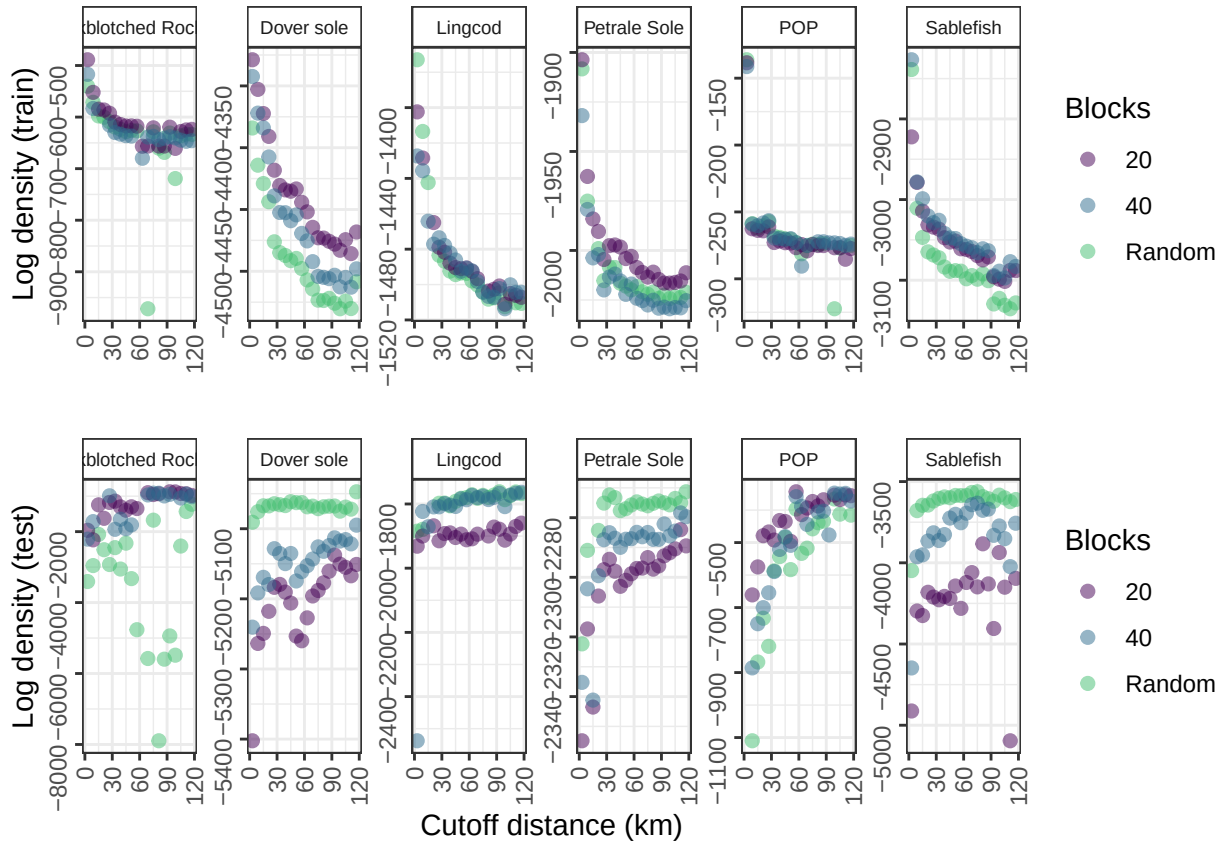


Figure 4: Gamma predictive densities for catch rate models for six species from the WCBTS survey, as a function of cutoff distances (larger cutoff distances correspond to coarser meshes). The in-sample predictive density is calculated holding out the first fold.

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